Preconditioning for quantum linear solvers in PDE-constrained optimization

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Abstract

Quantum linear system solvers cost a number of operations proportional to an *effective* condition number. This number is the ratio of the normalization with which the matrix is embedded in a unitary circuit (its block encoding) to its smallest singular value. Preconditioning is therefore the main lever that numerical linear algebra has on quantum solvers. The rules differ from the classical ones, however:

- clustering of eigenvalues does not help;
- preconditioners must be applied by multiplication, not by inner solves;
- every factor of a product must be well scaled.

We develop preconditioners under these rules for the saddle-point systems of PDE-constrained optimization. For distributed control of the Poisson equation, we combine a Schöberl–Zulehner block preconditioner with an additive multilevel (BPX) factor. We apply it on both sides of the KKT matrix and factor the result as $\mathcal{G}^* \mathcal{J} \mathcal{G}$, where $\mathcal{J}$ is a single two-qubit reflection and $\mathcal{G}$ is well scaled. A commuting property of quadrature factors and prolongations, with an element-local isometry, gives a block encoding of $\mathcal{G}$ whose normalization grows only like $\sqrt{\log h^{-1}}$. The effective condition number is then bounded independently of the mesh size. With level weights that we introduce, it is also bounded in the regularization parameter, up to a constant fixed by the mass matrix. For the time-periodic heat equation, a quantum Fourier transform in time decouples the problem into frequency-wise saddle-point problems of the same form. We prove a $\sqrt{3}$ bound uniform in all parameters, and the coupling matrix becomes a scaled reflection. Numerical experiments in one, two and three dimensions confirm the bounds. We also discuss when an asymptotic quantum advantage can and cannot be expected.

1 Introduction

Quantum linear system algorithms (QLSAs) prepare a quantum state proportional to the solution x of $Ax = b$. They use a number of calls to a circuit encoding A that scales like $\kappa \log(1/\varepsilon)$ [5, 6, 9]. The size n of the system enters only through the circuit cost, which is typically polylogarithmic in n for structured problems. Discretized PDEs are the obvious application, but the condition number of a discretized differential operator grows with the resolution, e.g. like h^{-2} for second-order elliptic problems. Without preconditioning, the dependence on κ cancels the gain in n [14, 16].

Preconditioning for QLSAs has been studied:

- by sparse approximate inverses [4];
- by "fast inversion" of a dominant part of the matrix [24];
- by wavelet bases [2];
- for elliptic finite element discretizations, by the BPX multilevel preconditioner [8];

- empirically, for CFD matrices, comparing several preconditioners and ways of encoding them [15];
- for the Newton systems of quantum interior point methods [20], which become increasingly ill conditioned along the central path.

These works make clear that preconditioning for a quantum solver is a different problem from classical preconditioning (Section 2). Only a bound on the condition number counts, not the distribution of the eigenvalues. The preconditioner must be realized as a circuit that *multiplies*. And the normalizations of all factors enter the cost.

Gap. The saddle-point systems of PDE-constrained optimization have not been considered from this angle. Classically, they are the prototype for parameter-robust block preconditioners [19, 21, 23], which are robust in the mesh size h and the regularization parameter β. Quantum approaches to PDE-constrained optimization have so far been variational heuristics.

Contributions.

1. We formulate design principles for preconditioners of QLSAs in matrix terms, centred on the effective condition number κ_α (Section 2).
2. For distributed control of the Poisson equation (Section 3), we combine the Schöberl–Zulehner block preconditioner with a diagonally scaled BPX factor R, applied on both sides.
   - The resulting matrix factors as $\mathcal{G}^*\mathcal{J}\mathcal{G}$. Here $\mathcal{J}$ is a single two-qubit reflection, and $\mathcal{G}$ maps multilevel coefficients to values and gradients at quadrature points.
   - A commuting lemma for quadrature factors and prolongations with an *element-local isometry* yields a block encoding of $\mathcal{G}$ with normalization $O(\sqrt{\log h^{-1}})$.
   - The effective condition number is bounded independently of h (Theorem 3.8). It holds for Q_k and P_k elements on nested meshes.
3. We introduce level weights that make the multilevel factor robust in β, up to the constant of the diagonally scaled mass matrix (Section 3.3).
4. For the time-periodic heat equation (Section 4), the QFT in time reduces the problem to frequency-wise problems of the Poisson type, with effective parameter $\beta_k = \beta/r_k^2$.
   - We prove a bound of $\sqrt{3}$ for the block preconditioner, uniform in h, τ, β and the frequency (Theorem 4.1).
   - The coupling matrix becomes a scaled reflection of norm at most one.
   - Neither the number of time steps nor the step size enters the effective condition number.
5. We verify all bounds and factorizations numerically in one, two and three dimensions (Section 6). We give an honest discussion of complexity: an asymptotic advantage over optimal classical solvers requires $d = 3$ for the elliptic problem. For the time-periodic problem it requires data that are rough in time (Section 7).

2 Quantum linear algebra in matrix terms

We describe the computational model in the language of numerical linear algebra (see [9, 13] for details).

States and outputs. A unit vector $v \in \mathbb{C}^{2^k}$ is stored in k qubits. Its entries cannot be read out. Repeated measurement gives estimates of quantities v^*Ov for suitable Hermitian O, with additive error ε. This costs $O(\varepsilon^{-2})$ repetitions, or $O(\varepsilon^{-1})$ with amplitude estimation [1]. The output of a quantum solver is therefore a small number of scalar *quantities of interest* (QoIs), such as $\langle w, y\rangle_M$ or the value of the objective, never the vector x.

Block encodings. A matrix A with $\|A\| \le \alpha$ is *block encoded* with *subnormalization* α if a unitary circuit U satisfies

$$U = \begin{bmatrix} A/\alpha & * \\ * & * \end{bmatrix}.$$

Applying U to $[v; 0]$ and keeping the first block (postselection) yields $Av/\|Av\|$. This succeeds with probability $\|Av\|^2/(\alpha\|v\|)^2$. Rectangular matrices are padded with zeros to square matrices of power-of-two size. Block encodings combine as follows (subnormalizations in brackets):

- products AB $[\alpha_A\alpha_B]$;
- Kronecker products $A \otimes B$ $[\alpha_A\alpha_B]$;
- linear combinations $\sum_j c_j A_j$ $[\sum_j |c_j|\alpha_j]$;
- stacking $[A_1; \ldots; A_k]$ $[(\sum_j \alpha_j^2)^{1/2}]$: prepare an index register with amplitudes $\propto \alpha_j$ and apply A_j controlled on index j;
- concatenation $[A_1, \ldots, A_k]$ $[\sqrt{k}\max_j \alpha_j]$;
- sparse matrices with s_r nonzeros per row and s_c per column, whose entries are computable by a circuit $[\sqrt{s_r s_c}\max_{ij} |a_{ij}|]$ [9].

In a product, the subnormalization can be far larger than the norm: $\alpha_A\alpha_B \ge \|A\|\|B\| \gg \|AB\|$ is possible.

Solving. Let $\mathcal{B}$ be Hermitian and block encoded with subnormalization α, and let its nonzero eigenvalues satisfy $|\lambda| \ge \sigma_{\min}$. Quantum singular value transformation (QSVT) [9] applies an odd polynomial $p(\mathcal{B}/\alpha)$ of degree D using D calls to the block encoding.

Why odd. QSVT alternates the encoding circuit U and its inverse U^* with single-qubit rotations. For $A = W\Sigma V^*$:

- an odd number of calls gives $Wp(\Sigma)V^*$ with p odd;
- an even number gives $Vp(\Sigma)V^*$ with p even.

This is the pattern of $A(A^*A)^k = W\Sigma^{2k+1}V^*$ versus $(A^*A)^k = V\Sigma^{2k}V^*$, as in Golub–Kahan bidiagonalization. U maps the right singular vectors into the block of the left ones, U^* maps them back, and the rotations in between only change the polynomial. Polynomials of mixed parity need a linear combination of both, which doubles α. For Hermitian $\mathcal{B} = Q\Lambda Q^*$ we have $W = Q\,\mathrm{sign}(\Lambda)$, $\Sigma = |\Lambda|$ and $V = Q$. An odd p therefore gives $Q\,\mathrm{sign}(\Lambda)p(|\Lambda|)Q^* = Qp(\Lambda)Q^*$, the usual matrix function with the signs of the eigenvalues kept. This is what an indefinite $\mathcal{B}$ requires.

With $p(t) \approx 1/t$ on $[\sigma_{\min}/\alpha, 1]$, with p bounded on $[-1, 1]$, we obtain the pseudoinverse $\mathcal{B}^+$, because $p(0) = 0$ for odd p. The cost is

$$O\big(\kappa_\alpha \log(1/\varepsilon)\big), \qquad \kappa_\alpha := \frac{\alpha}{\sigma_{\min}} \;\ge\; \kappa(\mathcal{B}), \tag{1}$$

which is optimal in κ_α [5, 6]. We call κ_α the *effective condition number*.

Design principles. Equation (1) changes what a good preconditioner is.

(D1) *Only κ_α counts.* QSVT does not adapt to the spectrum, unlike MINRES. Three distinct eigenvalues help MINRES but not a QLSA. What we need are inclusion intervals $[-b,-a]\cup[a,b]$ with b/a independent of the parameters, i.e. exactly the robustness results of [19,23].

(D2) *Subnormalizations count.* The preconditioned matrix must be encoded with α close to its norm. Multiplying separately encoded factors such as $P^{-1/2}$, $\mathcal{A}$ and $P^{-1/2}$ fails.

(D3) *Preconditioners multiply.* A classical inner solve (e.g. a multigrid cycle) has to become a circuit that multiplies by P^{-1} or by a factor of it. Additive multilevel methods [3,25] are sums of products of structured sparse matrices and fit this model. Multiplicative multigrid does not.

(D4) *Kernels are harmless.* QSVT inverts on the range, so redundant generating systems, such as multilevel ones, are allowed.

(D5) *Organize for QoIs.* Right-hand sides and outputs should be accessible without applying further badly scaled operators.

3 Distributed control of the Poisson equation

3.1 Problem and discretization

Let $\Omega=(0,1)^d$, $d\le 3$, with desired state $\hat{y}\in L^2(\Omega)$ and regularization parameter $\beta>0$. We solve

$$\min_{y,u}\ \tfrac{1}{2}\|y-\hat{y}\|^2_{L^2(\Omega)}+\tfrac{\beta}{2}\|u\|^2_{L^2(\Omega)}\quad\text{s.t.}\quad -\Delta y=u\text{ in }\Omega,\quad y=0\text{ on }\partial\Omega. \tag{2}$$

Grid and matrices. We use Q_1 elements on a uniform grid with $h=2^{-L}$. There are 2^L elements and 2^L-1 interior nodes per direction, so $n=(2^L-1)^d$ unknowns. In 1D, $M_1=\frac{h}{6}\,\mathrm{tridiag}(1,4,1)$ and $K_1=\frac{1}{h}\,\mathrm{tridiag}(-1,2,-1)$. In d dimensions the matrices are Kronecker products, e.g. $M=M_1\otimes M_1$ and $K=K_1\otimes M_1+M_1\otimes K_1$ for $d=2$. This structure matches the tensor-product structure of a qubit register.

Factorizations through quadrature points. Let $\xi_{1,2}=\frac{1}{2}\mp\frac{1}{2\sqrt{3}}$ be the two Gauss points on $[0,1]$. Let $F_0,F_1\in\mathbb{R}^{2^{L+1}\times(2^L-1)}$ map nodal values to $\sqrt{h/2}$ times the values, respectively derivatives, of the finite element function at the Gauss points of all elements. Two-point Gauss quadrature is exact for the integrands involved, so

$$M_1=F_0^TF_0,\qquad K_1=F_1^TF_1. \tag{3}$$

In d dimensions let $F_{\mathrm{val}}=F_0^{\otimes d}$ and let F_∇ stack the factors $F^{(i)}=F_0\otimes\cdots\otimes F_1\otimes\cdots\otimes F_0$ (F_1 in position i). Then $M=F_{\mathrm{val}}^TF_{\mathrm{val}}$ and $K=F_\nabla^TF_\nabla$. The factors have $2^{d(L+1)}$ rows, 2^d nonzeros per row and at most 4^d per column.

Remark 3.1 (Other elements). The construction is not specific to Q_1. Consider a nested family of conforming Lagrange spaces: Q_k on Cartesian grids, or P_k on uniformly red-refined, possibly unstructured simplicial meshes. Take a quadrature rule with positive weights that is exact for squares of the local polynomial space.

- The factorizations $M=F_{\mathrm{val}}^TF_{\mathrm{val}}$ and $K=F_\nabla^TF_\nabla$ are exact.
- The quadrature points are unisolvent, since $0<\int p^2=\sum_q w_q p(x_q)^2$.

- Lemma 3.6 below holds with an element-local isometry, the same for values and gradients, because derivatives stay in the local space.
- For Q_k the Kronecker form $M = M_1 \otimes M_1$, $K = K_1 \otimes M_1 + M_1 \otimes K_1$ ($d = 2$) holds for every k. The nodal basis functions are products $\phi_i(x)\phi_j(y)$ of 1D degree-k basis functions, so the integrals factor; only M_1 and K_1 change. For Q_2 they are 2-periodic (vertex and midpoint nodes alternate), with $2^{L+1} - 1$ unknowns per direction. Serendipity elements and P_k on simplices are not tensor products.
- Row counts that are not powers of two are zero-padded.
- Variable coefficients enter as one diagonal factor.

Mass lumping, curved elements and non-nested hierarchies keep the factorization only up to spectral equivalence. Unstructured meshes lose the Kronecker structure of the factors, which then need sparse-access oracles. We verified these statements numerically for Q_2 and Q_3 (1D and 2D) and for P_1 and P_2 on triangles. All factorization, commuting and isometry errors are below $3.3 \cdot 10^{-14}$.

Discrete problem and KKT system. Let $b = F_{\text{val}}^T \hat{y}_q$, where $\hat{y}_q$ holds the weighted values of $\hat{y}$ at the Gauss points. We minimize $\frac{1}{2} y^T M y - b^T y + \frac{\beta}{2} u^T M u$ subject to $Ky = Mu$. The optimality conditions are

$$\begin{bmatrix} M & 0 & K \\ 0 & \beta M & -M \\ K & -M & 0 \end{bmatrix} \begin{bmatrix} y \\ u \\ p \end{bmatrix} = \begin{bmatrix} b \\ 0 \\ 0 \end{bmatrix}. \tag{4}$$

We eliminate $u = p/\beta$, set $q = p/\sqrt{\beta}$ and scale the second equation. This gives

$$\mathcal{A}x = f, \qquad \mathcal{A} = \begin{bmatrix} M & \sqrt{\beta}K \\ \sqrt{\beta}K & -M \end{bmatrix}, \qquad x = \begin{bmatrix} y \\ q \end{bmatrix}, \quad f = \begin{bmatrix} b \\ 0 \end{bmatrix}, \tag{5}$$

with $u = q/\sqrt{\beta}$. The matrix $\mathcal{A}$ is symmetric and indefinite, with $\kappa(\mathcal{A}) = O(1 + \sqrt{\beta}h^{-2})$ (Table 1). Classically, MINRES with a robust block preconditioner and multigrid solves (5) in $O(n)$ operations [19, 21, 23].

3.2 A block preconditioner with a single elliptic operator

Let $A_\beta := M + \sqrt{\beta}K$, the stiffness matrix of $I - \sqrt{\beta}\Delta$.

Theorem 3.2 ([23]). *For $P = \text{blkdiag}(A_\beta, A_\beta)$, $\text{spec}(P^{-1}\mathcal{A}) \subset [-1, -1/\sqrt{2}] \cup [1/\sqrt{2}, 1]$ for all h and β.*

Proof. With $X^T M X = I$ and $X^T K X = \text{diag}(\lambda_j)$, the pencil $(\mathcal{A}, P)$ decouples into 2×2 pencils $\left(\begin{bmatrix} 1 & \mu_j \\ \mu_j & -1 \end{bmatrix}, (1 + \mu_j) I \right)$ with $\mu_j = \sqrt{\beta}\lambda_j > 0$. Their eigenvalues are $\pm\sqrt{1 + \mu_j^2}/(1 + \mu_j) \in \pm[1/\sqrt{2}, 1]$. □

Remark 3.3. In the Pearson–Wathen approximation of the Schur complement of (4) [19], $\widehat{S} = (K + M/\sqrt{\beta})M^{-1}(K + M/\sqrt{\beta})$, the operator to invert is again $K + M/\sqrt{\beta} = A_\beta/\sqrt{\beta}$. We use the form of Theorem 3.2 because it avoids M^{-1} and has the sharper interval.

3.3 A multilevel factor and level weights

For levels $\ell = 1, \dots, L$ with $h_\ell = 2^{-\ell}$, let $P_\ell \in \mathbb{R}^{n \times n_\ell}$ be the prolongation from level ℓ to level L ($P_L = I$). The level matrices $A_{\beta,\ell}$ have constant diagonal on a uniform grid:

$$\operatorname{diag}(A_{\beta,\ell}) = \delta_\ell I, \qquad \delta_\ell = \left(\tfrac{2h_\ell}{3}\right)^d + \sqrt{\beta}\,\tfrac{2d}{h_\ell}\left(\tfrac{2h_\ell}{3}\right)^{d-1}.$$

We define the weighted, diagonally scaled BPX factor

$$R = \left[\, \omega_1 P_1,\ \dots,\ \omega_L P_L \,\right], \qquad C = RR^T = \sum_{\ell=1}^{L} \omega_\ell^2 P_\ell P_\ell^T, \qquad \omega_\ell^2 = \frac{\theta_\ell}{\delta_\ell}, \tag{6}$$

with level weights

$$\theta_\ell = \min\left(1, \sqrt{\beta}\, h_\ell^{-2}\right)^s, \qquad s \in \{0, 1\}. \tag{7}$$

The choice $s = 0$ is plain BPX. Since $P_L = I$, R has full row rank.

Assumption 3.4. $\operatorname{spec}(CA_\beta) \subset [c_0, c_1]$, *with* c_1/c_0 *bounded independently of* h *and, ideally, of* β.

For fixed β and $s = 0$, this is the classical BPX theory for H^1-elliptic problems [3, 18, 25]. Robustness in β concerns the singularly perturbed operator $I - \sqrt{\beta}\Delta$.

- *Plain BPX* is robust in h but degrades as $\beta \to 0$ (Section 6): coarse levels on which the mass term dominates ($\sqrt{\beta} h_\ell^{-2} < 1$) all count the same smooth modes.
- *The weights* (7) with $s = 1$ damp these levels. They keep $\kappa(CA_\beta)$ small in all our tests, except when the grid does not resolve β at all ($h > \beta^{1/4}$). In that case $A_\beta \approx M$, and $\kappa(CA_\beta)$ approaches the constant 3^d of the Jacobi-scaled Q_1 mass matrix.
- *Truncation does not work.* Simply removing the mass-dominated levels makes things worse at intermediate β.

[TODO: Prove Assumption 3.4 for $s = 1$ with β-independent constants (robust additive multilevel methods for $I - \varepsilon^2\Delta$); a better mass factor than diagonal scaling would lower the 3^d floor.]

3.4 The two-sided system and its factorization

Let $\widetilde{R} = I_2 \otimes R$ and

$$\mathcal{B} := \widetilde{R}^T \mathcal{A} \widetilde{R} = \begin{bmatrix} R^T M R & \sqrt{\beta} R^T K R \\ \sqrt{\beta} R^T K R & -R^T M R \end{bmatrix}. \tag{8}$$

Theorem 3.5. *Under Assumption 3.4:*

1. $\mathcal{A}x = f$ *if and only if* $x = \widetilde{R} z$ *for a solution* z *of the consistent system* $\mathcal{B}z = \widetilde{R}^T f$. *In particular* $x = \widetilde{R}\mathcal{B}^+ \widetilde{R}^T f$.
2. *The nonzero eigenvalues of* $\mathcal{B}$ *lie in* $[-c_1, -c_0/\sqrt{2}] \cup [c_0/\sqrt{2}, c_1]$.

Proof. (1) $\widetilde{R}$ is surjective, so $\widetilde{R}^T$ is injective, and $\operatorname{range}(\mathcal{B}) = \widetilde{R}^T \mathcal{A} \mathbb{R}^{2n} = \operatorname{range}(\widetilde{R}^T)$. (2) The nonzero eigenvalues of $\widetilde{R}^T \mathcal{A} \widetilde{R}$ are those of $\widetilde{C}^{1/2} \mathcal{A} \widetilde{C}^{1/2} = S(P^{-1/2} \mathcal{A} P^{-1/2}) S^T$, where $\widetilde{C} = I_2 \otimes C$ and $S = \widetilde{C}^{1/2} P^{1/2}$. By Ostrowski's theorem [10, Thm. 4.5.9], each one equals $\theta\lambda$, where λ is an eigenvalue as in Theorem 3.2 and $\theta \in \operatorname{spec}(SS^T) = \operatorname{spec}(I_2 \otimes CA_\beta) \subset [c_0, c_1]$. □

Factorization with a reflection. *Why a factorization is needed.* Classically, a preconditioned operator $\widetilde{R}^T \mathcal{A} \widetilde{R}$ is applied factor by factor, and the norms of the factors are irrelevant. A Krylov method sees only the spectrum of the product. On a quantum computer, a product of block encodings encodes $\widetilde{R}^T \mathcal{A} \widetilde{R}/(\alpha_R^2 \alpha_{\mathcal{A}})$, because every factor is applied with a loss of amplitude. Its subnormalization satisfies

$$\alpha \;\geq\; \|R\|^2 \, \|\mathcal{A}\| \;=\; \|C\| \, \|\mathcal{A}\| \;\geq\; c_0 \|A_\beta^{-1}\| \cdot \tfrac{1}{2} \|A_\beta\| \;=\; \tfrac{c_0}{2} \, \kappa(A_\beta). \tag{9}$$

The steps are as follows.

- Each factor X has $\alpha_X \geq \|X\|$, since X/α_X is a block of a unitary matrix, and the α's multiply.
- $\|R\|^2 = \|RR^T\| = \|C\| \geq c_0 \|A_\beta^{-1}\|$ by Assumption 3.4.
- $\|A_\beta\| \leq \|M\| + \sqrt{\beta}\|K\| \leq 2\|\mathcal{A}\|$, because M and $\sqrt{\beta}K$ are blocks of $\mathcal{A}$.

Since $\sigma_{\min}(\mathcal{B}) \leq \|\mathcal{B}\| \leq c_1$, the effective condition number of the product encoding is at least $\frac{c_0}{2c_1}\kappa(A_\beta)$, with $\kappa(A_\beta) \sim 1 + \sqrt{\beta}h^{-2}$. Numerically this is close to $\kappa(\mathcal{A})$. For example, for $d = 1$, $L = 8$, $\beta = 1$: $\|R\|^2\|\mathcal{A}\| = 3.0 \cdot 10^4$, $\kappa(\mathcal{A}) = 2.6 \cdot 10^4$, while the factored form below gives $\|G\|^2/\sigma_{\min}(\mathcal{B}) = 5.0$. QSVT then has to invert on $[\sigma_{\min}/\alpha, 1]$, with degree $\sim \alpha/\sigma_{\min}$. It is non-adaptive: it behaves like Chebyshev semi-iteration with the upper eigenvalue bound overestimated as α instead of $\lambda_{\max}$. So the cost would be of the order of $\kappa(\mathcal{A})$, as without preconditioning. Krylov methods do not suffer from this, because they adapt to the actual spectrum. We therefore write $\mathcal{B}$ with factors that are well scaled individually.

Why a reflection in the middle. $\mathcal{B}$ is indefinite, so it cannot be written as $G^T G$. Let

$$G = \begin{bmatrix} F_{\text{val}} R \\ \beta^{1/4} F_\nabla R \end{bmatrix}, \qquad \mathcal{G} = I_2 \otimes G = \begin{bmatrix} G & 0 \\ 0 & G \end{bmatrix}, \qquad \mathcal{J} = \begin{bmatrix} E_{\text{val}} & E_\nabla \\ E_\nabla & -E_{\text{val}} \end{bmatrix},$$

where E_{val} and $E_\nabla = I - E_{\text{val}}$ are the diagonal 0/1 matrices that select the value rows and the gradient rows of G. Then

$$\mathcal{B} = \mathcal{G}^T \mathcal{J} \mathcal{G}, \qquad \mathcal{J} = \mathcal{J}^T = \mathcal{J}^{-1}, \qquad \|G\|^2 = \|R^T A_\beta R\| \leq c_1. \tag{10}$$

This is a J-factorization, as in LDL^T with $D = \pm 1$ or hyperbolic QR. G maps coefficients to values v and gradients g at the quadrature points, for both y and q. $\mathcal{J}$ is the matrix of the bilinear form

$$z^T \mathcal{B} z = \|v_y\|^2 - \|v_q\|^2 + 2\langle g_y, g_q \rangle.$$

On value rows it multiplies the y-block by $+1$ and the q-block by -1, giving $\pm R^T M R$. On gradient rows it swaps the two blocks, giving the off-diagonal $\sqrt{\beta} R^T K R$.

- $\mathcal{J}$ is a signed permutation matrix. On the device it is a single two-qubit gate acting on the qubit that labels the block (y or q) and the qubit that labels the row type (value or gradient): "if value row, flip the sign of the q-block; if gradient row, swap the blocks". Its subnormalization is 1.
- The right-hand side is $\widetilde{R}^T f = \mathcal{G}^T g$, where g is $\hat{y}_q$ placed in the value rows of the y-block (zero elsewhere).
- Outputs are accessed through $\mathcal{G}$ as well, since $G_{\text{val}} z_y = F_{\text{val}} y$. For example, $\langle w, y \rangle_M$ is an overlap of two states prepared with $\mathcal{G}$. So R is never applied on its own (D5).

3.5 Block encoding of the multilevel factor

A product encoding of $F_\nabla P_\ell$ loses a factor h_ℓ/h. The reason is that $\|F_\nabla\|$ is attained by oscillatory vectors and $\|P_\ell\|$ by smooth ones. Numerically, the loss reaches 148 for $L = 8$ (Table 4). The remedy is to differentiate on the coarse level and only then transfer to the fine level.

Lemma 3.6 (commuting with prolongation)**.** *For each level ℓ there is an isometry T_ℓ ($T_\ell^T T_\ell = I$) with*

$$F_{\mathrm{val}} P_\ell = T_\ell F_{\mathrm{val},\ell}, \qquad F^{(i)} P_\ell = T_\ell F_\ell^{(i)} \quad (i = 1, \dots, d).$$

T_ℓ is element-local and the same on every coarse element. In 1D, $T_\ell = I_{2^\ell} \otimes \widehat{T}_{L-\ell}$, where $\widehat{T}_k \in \mathbb{R}^{2^{k+1}\times 2}$ maps the two weighted Gauss values of an affine function on a coarse element to its weighted values at the 2^{k+1} fine Gauss points. In d dimensions, T_ℓ is the d-fold Kronecker product.

Proof. On a coarse element, coarse functions and their partial derivatives are affine in each direction. The two Gauss points are unisolvent for affine functions, which defines $\widehat{T}_k$. Both the coarse and the fine weighted values compute the exact L^2 norm of the same affine function, because two-point Gauss quadrature is exact for quadratics. Hence $\widehat{T}_k$ is an isometry. □

Some matrix T with $FP_\ell = TF_\ell$ always exists when F_ℓ has full column rank, e.g. $T = FP_\ell F_\ell^+$. The content of Lemma 3.6 is that T_ℓ is an isometry and element-local. This is what gives a cheap block encoding with subnormalization one. By the lemma,

$$G = \begin{bmatrix} T_1 G_1, & \dots, & T_L G_L \end{bmatrix}, \qquad G_\ell = \omega_\ell \begin{bmatrix} F_{\mathrm{val},\ell} \\ \beta^{1/4} F_\ell^{(1)} \\ \vdots \\ \beta^{1/4} F_\ell^{(d)} \end{bmatrix}, \qquad G_\ell^T G_\ell = \theta_\ell \delta_\ell^{-1} A_{\beta,\ell}. \tag{11}$$

Proposition 3.7. *G has a block encoding with subnormalization $\alpha_G \le C_d\sqrt{L}$, where C_d depends only on d.*

Sketch.

- G_ℓ has 2^d nonzeros per row and $(d+1)4^d$ per column.
- By the definition of δ_ℓ, its entries are bounded by 1, uniformly in h and β. The sparse-access encoding therefore gives $\alpha(G_\ell) \le (2^d(d+1)4^d)^{1/2}$.
- Each T_ℓ is an isometry, so it can be completed to a unitary.
- Concatenating the L levels costs a factor $\sqrt{L}$.

□

The sparse-access constant is pessimistic ($C_3 \approx 45$). Encoding G_ℓ as a weighted stack of Kronecker products of 1D two-term factors gives constants close to $\|G_\ell\| \le \sqrt{\theta_\ell}\, 3^{d/2}$. **[TODO: compute α_G for the explicit construction; can the $\sqrt{L}$ be avoided?]**

Theorem 3.8. *Under Assumption 3.4, QSVT applied to $\mathcal{B} = \mathcal{G}^T \mathcal{J} \mathcal{G}$ solves $\mathcal{B}z = \mathcal{G}^T g$ to accuracy ε using*

$$O\Big(\frac{C_d^2 L}{c_0} \log \frac{1}{\varepsilon}\Big)$$

calls to block encodings of G, G^T and $\mathcal{J}$. Each call costs poly(L) *gates. The count depends on h only through $L = \log_2 h^{-1}$, and on β only through c_0.*

Proof. By (10), Theorem 3.5 and Proposition 3.7, $\kappa_\alpha(\mathcal{B}) \le \sqrt{2}\,\alpha_G^2/c_0 \le \sqrt{2}\,C_d^2 L/c_0$. The result follows from (1). □

[TODO: End-to-end statement: preparation of g, amplitude amplification of the success probability, estimation of $\|z\|$, and readout of the QoI.]

4 The time-periodic heat equation

4.1 Problem, discretization and Fourier transform in time

We solve

$$\min_{y,u} \frac{1}{2}\int_0^T \|y-\hat{y}\|^2_{L^2(\Omega)}dt + \frac{\beta}{2}\int_0^T \|u\|^2_{L^2(\Omega)}dt \quad \text{s.t.} \quad \partial_t y - \Delta y = u, \quad y|_{\partial\Omega}=0, \quad y(\cdot,0)=y(\cdot,T). \tag{12}$$

Classically, this problem is treated with multiharmonic finite elements and robust block preconditioners [11, 12].

- *Space:* Q_1 as in Section 3.
- *Time:* $N_t = 2^{L_t}$ steps of length $\tau = T/N_t$ with periodic backward Euler, $M(y_k - y_{k-1})/\tau + Ky_k = Mu_k$, with $y_0 = y_{N_t}$.

With the cyclic shift Σ and $D_t = (I-\Sigma)/\tau$, the constraint reads

$$\mathcal{K}y = \mathbb{M}u, \qquad \mathcal{K} = D_t \otimes M + I \otimes K, \qquad \mathbb{M} = I \otimes M.$$

We approximate the objective by the rectangle rule. As in (5), we eliminate u and set $q = p/(\tau\sqrt{\beta})$. This gives the symmetric system

$$\mathcal{A} = \begin{bmatrix} \mathbb{M} & \sqrt{\beta}\,\mathcal{K}^T \\ \sqrt{\beta}\,\mathcal{K} & -\mathbb{M} \end{bmatrix}, \qquad x = \begin{bmatrix} y \\ q \end{bmatrix}, \qquad f = \begin{bmatrix} \mathbb{M}\hat{y} \\ 0 \end{bmatrix}, \qquad u = \frac{q}{\sqrt{\beta}}. \tag{13}$$

D_t is circulant: $D_t = \mathcal{F}^*\Lambda\mathcal{F}$, with the unitary DFT matrix $\mathcal{F}$ and

$$\lambda_k = \frac{1-e^{-2\pi ik/N_t}}{\tau} = \frac{2}{\tau}\sin\Big(\frac{\pi k}{N_t}\Big)e^{\,i(\pi/2-\pi k/N_t)}, \qquad \operatorname{Re}\lambda_k \geq 0. \tag{14}$$

Applying $\mathcal{F}$ to the time index block-diagonalizes $\mathcal{A}$ into the Hermitian blocks

$$\mathcal{A}_k = \begin{bmatrix} M & \sqrt{\beta}(K+\bar{\lambda}_k M) \\ \sqrt{\beta}(K+\lambda_k M) & -M \end{bmatrix}, \qquad k = 0, \ldots, N_t - 1. \tag{15}$$

On a quantum computer, $\mathcal{F}$ is the QFT on the L_t time qubits, with $O(L_t^2)$ gates. It is unitary, so it does not change the condition number or the subnormalization. The analysis below uses only $\operatorname{Re}\lambda_k \geq 0$. It therefore also covers multiharmonic discretizations, where $\lambda_k = i\omega_k$.

4.2 A robust preconditioner per frequency

Theorem 4.1. *Let $r_k = 1 + \sqrt{\beta}|\lambda_k|$, $A_k = r_k M + \sqrt{\beta}K$ and $P_k = \operatorname{blkdiag}(A_k, A_k)$. Then for all h, τ, β and k,*

$$\operatorname{spec}(P_k^{-1}\mathcal{A}_k) \subset [-1, -1/\sqrt{3}] \cup [1/\sqrt{3}, 1].$$

Proof. With $X^T M X = I$ and $X^T K X = \operatorname{diag}(\mu_j)$, the pencil decouples into 2×2 pencils with eigenvalues $\pm\sqrt{1+x^2}/(1+c)$, where $x = \sqrt{\beta}|\mu_j + \lambda_k|$ and $c = \sqrt{\beta}(\mu_j + |\lambda_k|)$.

- $c \geq x$, by the triangle inequality.
- $c \leq \sqrt{2}\,x$, because $\mu_j + |\lambda_k| \leq (\mu_j + \operatorname{Re}\lambda_k) + |\operatorname{Im}\lambda_k| \leq \sqrt{2}|\mu_j + \lambda_k|$ when $\operatorname{Re}\lambda_k \geq 0$.

Hence $\sqrt{1+x^2}/(1+\sqrt{2}x) \leq \sqrt{1+x^2}/(1+c) \leq 1$. The lower bound is minimal at $x = \sqrt{2}$, where it equals $1/\sqrt{3}$. □

Remark 4.2.

- *Time domain.* In the time domain the preconditioner is $\operatorname{blkdiag}(\mathcal{P}_t, \mathcal{P}_t)$ with $\mathcal{P}_t = I \otimes (M + \sqrt{\beta}K) + \sqrt{\beta}|D_t| \otimes M$ and $|D_t| = (D_t^T D_t)^{1/2}$. It is real and symmetric, although $\mathcal{K}$ is not. $|D_t|$ is the discrete analogue of $|\partial_t|$, which underlies the $H^{1/2}$-in-time norms of parabolic theory.
- *Relation to prior work.* Theorem 4.1 is the analogue, for this discretization, of the robust multiharmonic preconditioners of [11,12]. **[TODO: compare constants and attribute.]**

4.3 Each frequency is a Poisson problem

We can write $A_k = r_k(M + \sqrt{\beta_k}K)$ with $\sqrt{\beta_k} = \sqrt{\beta}/r_k$. Each frequency is therefore the operator of Section 3 with effective parameter β_k, scaled by r_k.

- *Multilevel factor.* $R_k = [\omega_{1,k}P_1, \ldots, \omega_{L,k}P_L]$ with $\omega_{\ell,k}^2 = \theta_{\ell,k}/\delta_{\ell,k}$, where $\delta_{\ell,k} = r_k(\frac{2h_\ell}{3})^d + \sqrt{\beta}\frac{2d}{h_\ell}(\frac{2h_\ell}{3})^{d-1}$ and $\theta_{\ell,k} = \min(1, \sqrt{\beta_k}h_\ell^{-2})^s$.
- *Frequency enters only through scalars.* The prolongations P_ℓ and isometries T_ℓ are independent of k.
- *The level weights are no longer optional.* β_k decreases from β ($k = 0$) to $\beta\tau^2/(\tau + 2\sqrt{\beta})^2$ ($k = N_t/2$), so high frequencies are mass-dominated.

Factorization with a scaled reflection. Let

$$G_k = \begin{bmatrix} \sqrt{r_k}\, F_{\mathrm{val}} R_k \\ \beta^{1/4} F_\nabla R_k \end{bmatrix}, \qquad G_k^* G_k = R_k^T A_k R_k.$$

Then $\mathcal{B}_k := \widetilde{R}_k^T \mathcal{A}_k \widetilde{R}_k = \mathcal{G}_k^* \mathcal{J}_k \mathcal{G}_k$, where

$$\mathcal{J}_k = \begin{bmatrix} j_{11}E_{\mathrm{val}} & j_{12}E_{\mathrm{val}} + E_\nabla \\ j_{21}E_{\mathrm{val}} + E_\nabla & j_{22}E_{\mathrm{val}} \end{bmatrix}, \qquad J_k = \begin{bmatrix} j_{11} & j_{12} \\ j_{21} & j_{22} \end{bmatrix} = \frac{1}{r_k}\begin{bmatrix} 1 & \sqrt{\beta}\bar{\lambda}_k \\ \sqrt{\beta}\lambda_k & -1 \end{bmatrix} = s_k U_k. \quad (16)$$

On value rows, $\mathcal{J}_k$ applies the 2×2 matrix J_k across the two blocks. On gradient rows, it swaps the blocks as before. U_k is a Hermitian unitary 2×2 matrix (a reflection) with phase $\arg \lambda_k = \pi/2 - \pi k/N_t$, and

$$s_k = \frac{\sqrt{1 + \beta|\lambda_k|^2}}{1 + \sqrt{\beta}|\lambda_k|} \in \left[\tfrac{1}{\sqrt{2}}, 1\right].$$

Hence $\|\mathcal{J}_k\| \le 1$, and $\mathcal{J}_k$ is block encoded with subnormalization 1. The weight $\sqrt{r_k}$ on the value rows is essential: without it, $\|J_k\|$ would grow like $\sqrt{\beta}/\tau$.

Theorem 4.3. *Suppose Assumption 3.4 holds for all A_k with constants c_0, c_1 independent of k. Then the nonzero eigenvalues of $\mathcal{B} = \mathrm{blkdiag}_k(\mathcal{B}_k)$ lie in $[-c_1, -c_0/\sqrt{3}] \cup [c_0/\sqrt{3}, c_1]$, and $\kappa_\alpha(\mathcal{B}) \le \sqrt{3}\,\alpha_G^2/c_0$ with $\alpha_G \le C_d\sqrt{L}$. In particular, neither N_t nor τ enters κ_α.*

Proof. Argue as for Theorem 3.5 and Proposition 3.7, frequency by frequency, with Theorem 4.1 in place of Theorem 3.2. □

QSVT applies $\mathcal{B}^+$ to all frequencies simultaneously, in superposition. Compared with the Poisson problem, the cost per solve increases only by two QFTs and the frequency-controlled scalars (Section 5).

5 Implementation

Registers.

- $d(L + 1)$ qubits for quadrature rows (element index and Gauss point per direction).
- dL qubits for nodal unknowns. Level-ℓ data use the leading ℓ bits per direction. The unused index $2^\ell - 1$, a boundary node, is mapped to zero by a comparator flag.
- $\lceil \log_2 L \rceil$ qubits for the level, $\lceil \log_2(d + 1) \rceil$ for the component c, one qubit b for the block, and L_t qubits for time or frequency.
- Ancillas for the block encodings and QSVT.

Coarse element indices are the leading bits of fine element indices, so $T_\ell = I \otimes \widehat{T}_{L-\ell}$ acts on trailing bits only. Unused index combinations lie in the kernel of G, which is harmless by (D4).

Circuits.

1. *1D two-term factors* $F_{0,\ell}$ and $F_{1,\ell}$: a shift by one on ℓ qubits ($O(\ell)$ gates), an LCU of two diagonal patterns, and the boundary flag. Kronecker products and a weighted stack over c then give G_ℓ.
2. *The isometry* $\widehat{T}_k$: its two orthonormal columns are the uniform vector (Hadamard gates) and a normalized discrete ramp with Gauss offsets. **[TODO: reference for linear-amplitude state preparation.]**
3. *Levels*: G_ℓ and T_ℓ controlled on the level register, followed by Hadamard gates and post-selection on the level register.
4. $\mathcal{J}$: one two-qubit gate (Poisson). For the heat equation, $\mathcal{J}_k$ needs U_k controlled on c and k, plus one ancilla rotation for s_k.
5. *QSVT phase factors* for an odd polynomial approximating $1/t$ on $[\sigma_{\min}/\alpha, 1]$, of degree $O(\kappa_\alpha \log(1/\varepsilon))$. They are computed classically, e.g. as in [7], using a lower bound for c_0.
6. *Frequency scalars (heat).* $\arg \lambda_k$ is affine in k, so it needs only single-qubit phase gates. The magnitude $|\lambda_k| = \frac{2}{\tau}\sin(\pi k/N_t)$ is an LCU of $\mathrm{diag}(e^{\pm i\pi k/N_t})$. The derived quantities $r_k^{\pm 1/2}$, $\omega_{\ell,k}$ and s_k come from QSVT polynomials or reversible arithmetic. **[TODO: compare costs.]**
7. *Right-hand side and readout.* Prepare g (weighted values of $\hat{y}$ at Gauss points, and time steps for the heat problem); for smooth $\hat{y}$, efficient circuits exist **[TODO: refs]**. Then apply $\mathcal{G}^T$, after a QFT in time for the heat problem. QoIs are overlaps $\langle \mathcal{G}g_w, \mathcal{G}z\rangle$, estimated by Hadamard tests and amplitude estimation; $\|z\|$ follows from the success probability.

Classical precomputation. All weights are in closed form. The only nontrivial precomputation is the QSVT phase factors and the constant c_0 that enters them.

6 Numerical experiments

All experiments use dense or sparse linear algebra in double precision, with scripts in the accompanying repository:

- `qp_poisson_check.py`: Table 1, Table 4;
- `qp_bpx_weights.py`: Table 2;
- `qp_bpx_sparse.py`: Table 3, extreme eigenvalues by Lanczos with CG-based inner solves;
- `qp_heat_periodic_check.py`: Table 5.

Condition numbers on the range are ratios of the largest to the smallest nonzero eigenvalue magnitudes.

Poisson control: preconditioner and factorization. Table 1 shows:

- $\kappa(\mathcal{A})$ grows like $\sqrt{\beta}h^{-2}$;
- the interval of Theorem 3.2 is attained;
- $\kappa(\mathcal{B})$ on the range follows $\kappa(CA_\beta)$, as predicted by Theorem 3.5;
- $\|G\|$ is $O(1)$;
- the factorization $\mathcal{B} = \mathcal{G}^T\mathcal{J}\mathcal{G}$ holds to machine precision in all cases.

d	L	β	$\kappa(\mathcal{A})$	$\lvert\operatorname{spec}(P^{-1}\mathcal{A})\rvert$	$\kappa(CA_\beta)$	$\kappa(\mathcal{B})$	$\lVert G\rVert$
1	4	1	$1.0 \cdot 10^2$	$[0.913, 1.000]$	3.45	3.49	1.86
1	6	1	$1.7 \cdot 10^3$	$[0.913, 1.000]$	4.36	4.38	2.09
1	8	1	$2.6 \cdot 10^4$	$[0.913, 1.000]$	4.98	4.99	2.23
1	8	10^{-4}	$2.6 \cdot 10^3$	$[0.708, 1.000]$	4.43	4.50	2.11
1	8	10^{-8}	$2.6 \cdot 10^1$	$[0.707, 0.999]$	7.31	8.03	2.70
2	5	1	$2.1 \cdot 10^2$	$[0.953, 1.000]$	4.06	4.04	1.75
2	5	10^{-4}	$4.0 \cdot 10^1$	$[0.707, 0.996]$	3.96	3.47	1.74
2	5	10^{-6}	$4.2 \cdot 10^0$	$[0.707, 0.981]$	6.40	7.90	2.34

Table 1: Poisson control with plain BPX ($s = 0$). $\kappa(\mathcal{B})$ is computed on the range.

Robustness in h and β, including 3D. Tables 2 and 3 compare plain and weighted BPX.

- For fixed β, $\kappa(CA_\beta)$ is bounded in h in all dimensions. It levels off slowly (3D, $\beta = 1$: 4.2, 5.3, 6.1 for $L = 3, 4, 5$).
- Plain BPX degrades as $\beta \to 0$, up to 44.6 in 3D.
- The weights (7) with $s = 1$ keep $\kappa(CA_\beta) \le 6.2$ whenever the grid resolves β ($h \lesssim \beta^{1/4}$). In unresolved cases they approach the Jacobi constant 3^d of the mass matrix, with 20.5 as the worst value in 3D.

		β					
d, L	s	1	10^{-2}	10^{-4}	10^{-6}	10^{-8}	10^{-10}
1, 10	0	5.42	5.32	5.02	4.92	7.33	9.80
1, 10	1/2	5.42	5.30	4.97	4.55	3.88	2.86
1, 10	1	5.42	5.33	6.14	5.37	4.92	3.61
2, 4	0	3.58	3.49	3.62	7.23	18.89	26.31
2, 4	1/2	3.58	3.33	2.29	3.12	9.84	14.14
2, 4	1	3.58	3.29	2.26	1.97	7.16	10.51

Table 2: $\kappa(CA_\beta)$ for the level weights (7), including the intermediate exponent $s = \frac{1}{2}$. In 2D with $L = 4$, the grid does not resolve $\beta \le 10^{-8}$.

d	L	n	s	$\beta = 1$	10^{-2}	10^{-4}	10^{-6}	10^{-8}
2	6	3969	0	4.44	4.36	4.22	7.14	11.27
2	6	3969	1	4.44	4.27	3.53	2.83	1.64
3	3	343	0	4.16	4.23	5.95	17.36	35.95
3	3	343	1	4.16	3.71	2.77	8.80	20.47
3	4	3375	0	5.30	5.31	6.41	15.75	44.55
3	4	3375	1	5.30	4.90	3.39	4.26	17.69
3	5	29791	0	6.08	6.05	6.65	14.19	–
3	5	29791	1	6.08	5.74	4.46	3.12	–

Table 3: $\kappa(CA_\beta)$ on larger grids, plain ($s = 0$) and weighted ($s = 1$) BPX, computed by Lanczos.

Subnormalization loss of the naive product. Table 4 gives the ratio $\lVert F_1\rVert\lVert P_\ell\rVert/\lVert F_1 P_\ell\rVert$. A product encoding of $F_1 P_\ell$ loses this factor. It equals $2^{L-\ell}$, up to a correction on the coarsest levels. Lemma 3.6 removes it completely: the commuted factors satisfy $\lVert T_\ell\rVert\,\lVert F_{1,\ell}\rVert = \lVert F_1 P_\ell\rVert$.

ℓ	1	2	3	4	5	6	7
$\|F_1\|\|P_\ell\|/\|F_1P_\ell\|$	147.8	65.8	32.2	16.0	8.0	4.0	2.0

Table 4: Loss of the naive product encoding, $d = 1$, $L = 8$.

Time-periodic heat control. Table 5 shows:

- the global eigenvalues of $\mathcal{P}^{-1}\mathcal{A}$ lie in the interval $[1/\sqrt{3}, 1]$ of Theorem 4.1;
- per frequency, the factorization $\mathcal{B}_k = \mathcal{G}_k^*\mathcal{J}_k\mathcal{G}_k$ holds and $\|J_k\| \le 1$;
- the per-frequency BPX constants behave as in the Poisson case with effective parameters β_k.

d	L	N_t	β	$\kappa(\mathcal{A})$	$\lvert\operatorname{spec}(\mathcal{P}^{-1}\mathcal{A})\rvert$
1	5	32	1	$4.1 \cdot 10^2$	$[0.714, 1.000]$
1	5	32	10^{-2}	$2.9 \cdot 10^2$	$[0.603, 0.999]$
1	5	32	10^{-4}	$4.1 \cdot 10^1$	$[0.666, 0.992]$
1	5	32	10^{-6}	$4.2 \cdot 10^0$	$[0.702, 0.990]$
2	3	16	1	$1.3 \cdot 10^1$	$[0.811, 0.999]$
2	3	16	10^{-2}	$1.2 \cdot 10^1$	$[0.669, 0.993]$
2	3	16	10^{-4}	$2.7 \cdot 10^0$	$[0.684, 0.936]$
2	3	16	10^{-6}	$4.4 \cdot 10^0$	$[0.704, 0.981]$
[TODO: per-frequency BPX and factorization checks]					

Table 5: Time-periodic heat control. The bound of Theorem 4.1 is $1/\sqrt{3} \approx 0.577$.

7 Complexity and discussion

Poisson control. Suppose the QoI is needed to accuracy ε and the discretization error is $O(h^2)$, so $h \sim \varepsilon^{1/2}$.

- Classically, the cost is $O(n) = O(\varepsilon^{-d/2})$ with optimal preconditioning.
- On a quantum computer, Theorem 3.8 gives $\operatorname{polylog}(1/\varepsilon)$ per solve. Readout adds a factor $O(\varepsilon^{-1})$, for a total of $\widetilde{O}(\varepsilon^{-1})$.

An asymptotic advantage therefore requires $d = 3$ on this count. Deiml and Peterseim state an advantage from $d = 2$ on [8]. **[TODO: reconcile cost models.]**

Heat control.

- With backward Euler, $N_t \sim \varepsilon^{-1}$ and the classical cost is $\varepsilon^{-(d/2+1)}$. On that count, a quantum advantage would appear for every d.
- This comparison is unfair. For data that are smooth in time, a multiharmonic discretization needs only $\log(1/\varepsilon)$ harmonics, and the elliptic count returns.
- The time dimension favours the quantum solver only when many harmonics are genuinely needed, i.e. for targets or forcing that are rough in time. **[TODO: quantify.]**

Limitations.

- The algorithms need fault-tolerant hardware.
- The constants (C_d, the 3^d floor, readout) matter in practice.
- State preparation for general data and the end-to-end success probability need a full analysis.
- The robustness of the multilevel factor in β is so far supported numerically, not proven.

8 Conclusions and outlook

Preconditioning for quantum linear solvers is a numerical linear algebra problem with its own rules. Respecting them leads to factorized, well-scaled preconditioned operators whose effective condition numbers are robust in the discretization parameters. Open directions:

1. a proof of β-robust multilevel bounds, and a better mass factor;
2. removing the $\sqrt{L}$ factor and computing explicit constants;
3. control constraints, where semismooth Newton or interior point methods change the $(2,2)$ block in every iteration. This links to preconditioners for interior point methods and to quantum interior point methods [20];
4. initial value problems. There D_t is not circulant. The α-circulant preconditioner [17] needs the non-unitary scaling $\Gamma_\alpha = \mathrm{diag}(\alpha^{k/N_t})$, whose condition number $1/\alpha$ enters the subnormalization. This trade-off between preconditioner quality and encoding cost has no classical analogue. Wavelet-in-time bases [22] are an alternative;
5. variable coefficients, boundary control, and non-Cartesian meshes.

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